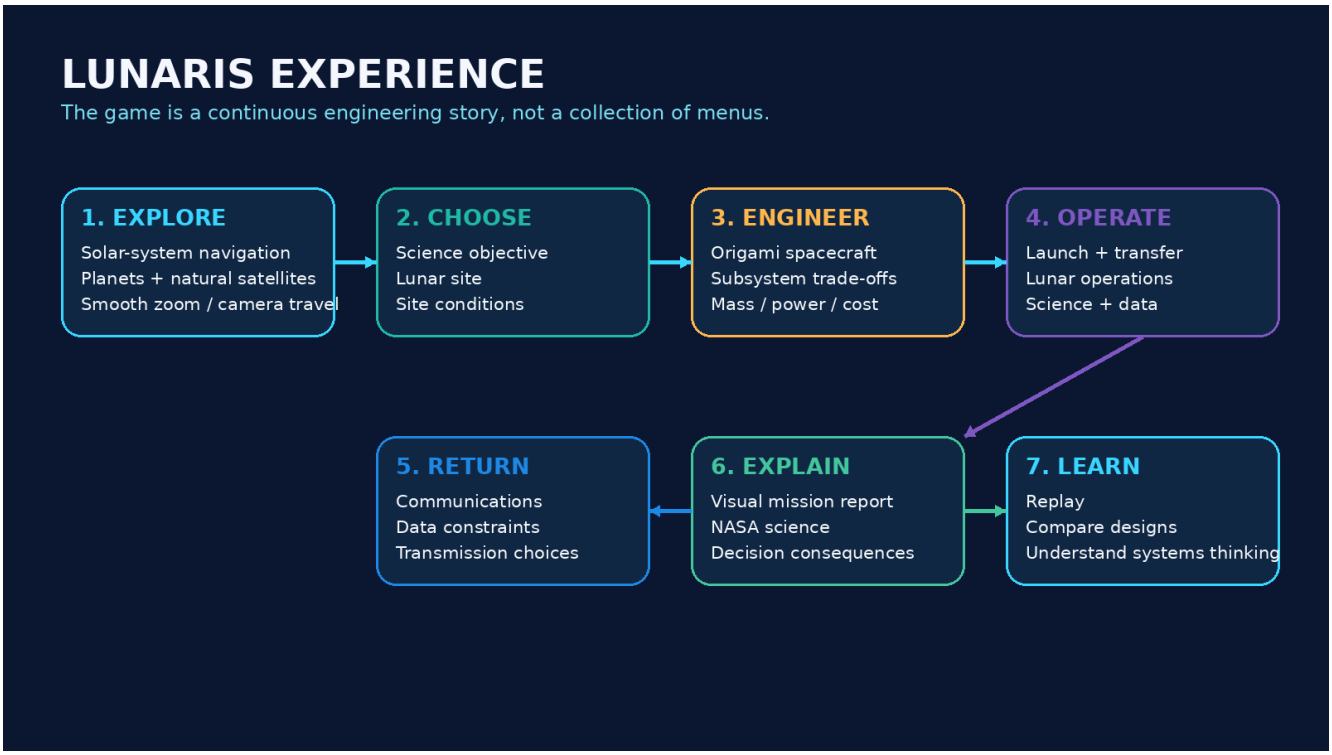
LUNARIS

DESIGN BEYOND EARTH

GAME DESIGN DOCUMENT




Explore the solar system. Choose the mission. Engineer the spacecraft. Discover the consequences.

NASA Space Apps Challenge 2026 | Challenge: Space Mission Design Game

# Executive Vision

A high-fidelity 2D lunar mission-engineering experience built as an educational systems sandbox.

LUNARIS is a browser-based game in which the player explores a living solar system, selects a lunar science objective and destination, engineers a specialized spacecraft, simulates the mission, and receives a visual scientific debrief explaining why the mission succeeded or failed.

**CORE HOOK**

The Moon does not adapt to your spacecraft. Your spacecraft must adapt to the Moon.

| Pillar | What the player does | Why it matters |
|---|---|---|
| Explore | Navigate planets and natural satellites with smooth camera travel. | Makes space discovery the opening experience. |
| Choose | Define what to learn and where to go on the Moon. | Creates mission requirements before engineering. |
| Engineer | Configure a specialized spacecraft through an Origami-inspired interaction. | Turns systems engineering into gameplay. |
| Operate | Manage orbit, power, data, communication, time and mission events. | Shows trade-offs propagating through the mission. |
| Understand | Read a visual mission/science report. | Makes the science understandable to non-technical players. |

# Challenge Alignment & Real-World Problem

The game is not merely a space-themed game; it is a simplified environment for integrated mission architecture.

The NASA challenge asks players to experience mission objectives, spacecraft design, scientific instruments, launch vehicles, budgets, power, mass, communications, data, mission lifetime and orbital constraints. LUNARIS treats these as connected systems rather than isolated stats.

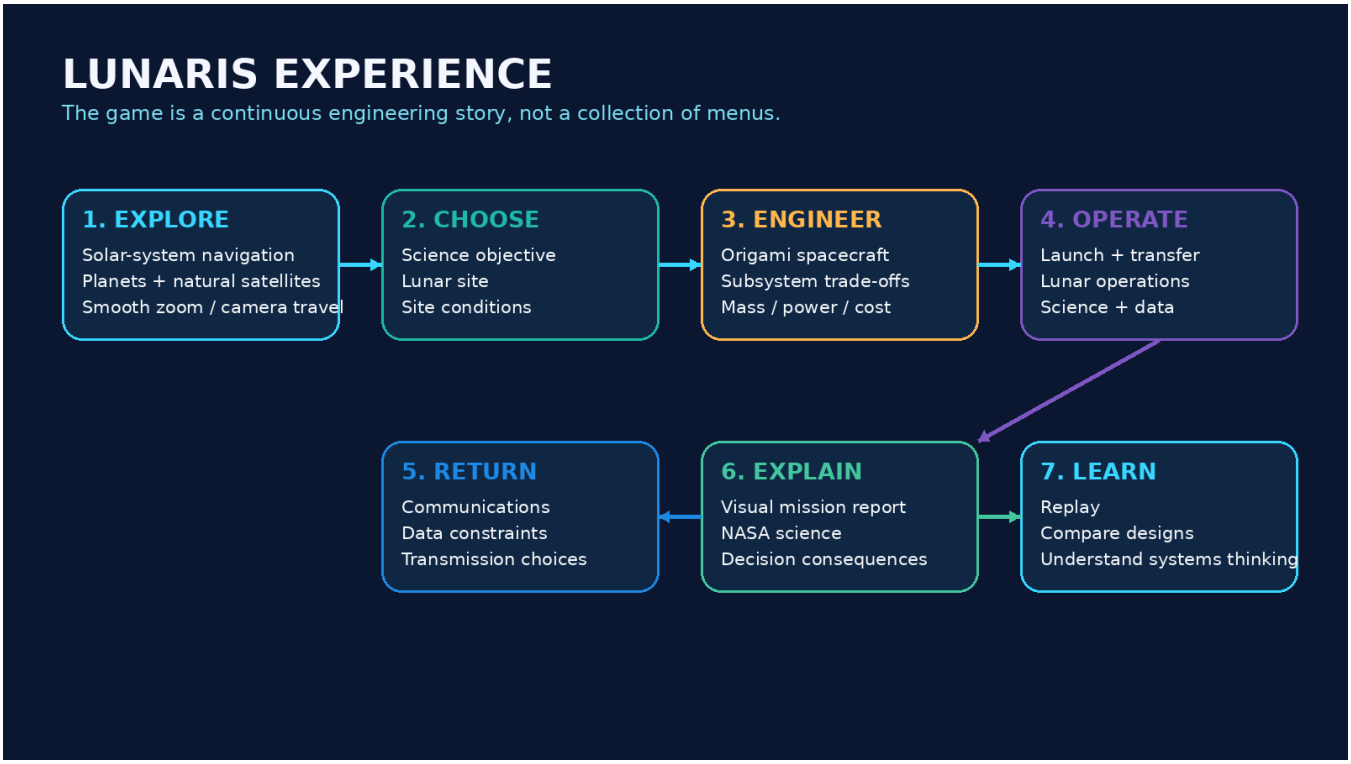
**REAL-WORLD PROBLEM**

Students and non-specialists rarely have an environment in which they can experiment with an entire mission and immediately see how a decision in one subsystem creates consequences elsewhere.

| Real mission problem | LUNARIS interpretation |
|---|---|
| Choose a scientifically valuable destination that is actually reachable. | Compare lunar locations and their environmental/communications/science profiles. |
| Translate mission objectives into engineering requirements. | Objective -> requirements -> spacecraft architecture. |
| Balance capability against mass, power, budget and time. | Each spacecraft subsystem has benefits and costs. |
| Return useful science through limited communications. | Instruments generate data; comms determine what can return. |
| Explain mission success or failure. | Visual post-mission analysis maps decisions to outcomes. |

# Player Journey

The experience is continuous: the camera, spacecraft, data and mission state evolve rather than jumping between disconnected menus.




The player should feel that the entire game is one connected mission story. A click on a planet begins a camera journey; a design choice physically changes the spacecraft; a mission event changes the visible telemetry; and the final report reconstructs the decisions as a visual explanation.

**DESIGN RULE**

Every major interaction should answer three questions visually: What changed? Why did it change? What does it affect next?

# Solar-System & Lunar Exploration Layer

The game opens as an interactive astronomical environment, not a conventional menu.

| Interaction | Visual behavior | Purpose |
|---|---|---|
| Hover / focus | Subtle illumination, label, orbital emphasis. | Helps players understand what is selectable. |
| Click planet | Camera eases toward the body and zooms smoothly. | Creates a sense of travelling through space. |
| Click natural satellite | Camera dives further into the object. | Creates a hierarchy: solar system -> planet -> moon. |
| Open info | Scientific information appears as an overlay around the object. | Makes the exploration educational. |
| Enter mission | Camera transitions from Moon exploration into the lunar mission layer. | Connects discovery to gameplay. |

Rendering should maintain a cinematic, high-fidelity 2D look: realistic-feeling lighting, atmospheric glows, layered planetary surfaces, smooth camera motion, subtle particles, and clear scientific overlays. It must not become pixel art, 8-bit, CRT or arcade-retro styling.

# Mission Objective & Lunar Site Selection

The player first decides what scientific question they are trying to answer, then chooses where on the Moon to answer it.

A mission objective creates requirements. A site creates environmental constraints. Those two inputs drive the spacecraft design.

| Objective examples | What it values | Likely spacecraft pressure |
|---|---|---|
| Surface mapping | Imaging coverage / terrain visibility | Camera, data storage, power, communications |
| Composition analysis | Scientific composition data | Spectrometer, power, data volume, downlink |
| Radiation / environment study | Environmental measurements | Radiation sensor, thermal management, data |
| Water / volatile investigation | Science at high-value targets | Target selection, payload, power, data, risk |

Candidate lunar sites should expose differences in terrain, illumination, temperature, communication conditions, science potential and resource/science relevance. The player is not selecting a location because it looks interesting; they are selecting a site whose conditions shape the mission architecture.

**CORE QUESTION**

What spacecraft architecture is appropriate for this particular place and this particular scientific objective?

# Specialized Spacecraft Engineering

The spacecraft is not a menu of cosmetic parts. It is the main engineering puzzle.

| Subsystem | Player controls | Primary trade-offs |
|---|---|---|
| Propulsion | Transfer / descent capability | Fuel, mass, cost, mission margin |
| Power | Generation + storage architecture | Mass, cost, endurance, instrument availability |
| Thermal | Protection level | Mass, cost, environmental survival |
| Communications | Antenna / transmission capability | Mass, power, bandwidth, data return |
| Navigation | Precision / terrain capability | Power, computation, cost, landing risk |
| Science payload | Instrument selection | Mass, power, cost, data, science value |
| Data system | Storage / handling capacity | Mass, power, useful data return |

The system is intentionally specialized: a spacecraft optimized for one lunar site may be a poor spacecraft for another. This is the core systems-engineering lesson.

# Origami-Inspired Spacecraft Interface

Origami is the interaction metaphor for configuration and deployment, not a childish art style.

The spacecraft starts as a compact geometric form. Engineering choices progressively unfold and configure its systems. Each fold/unfold operation is a visual representation of a real design decision.

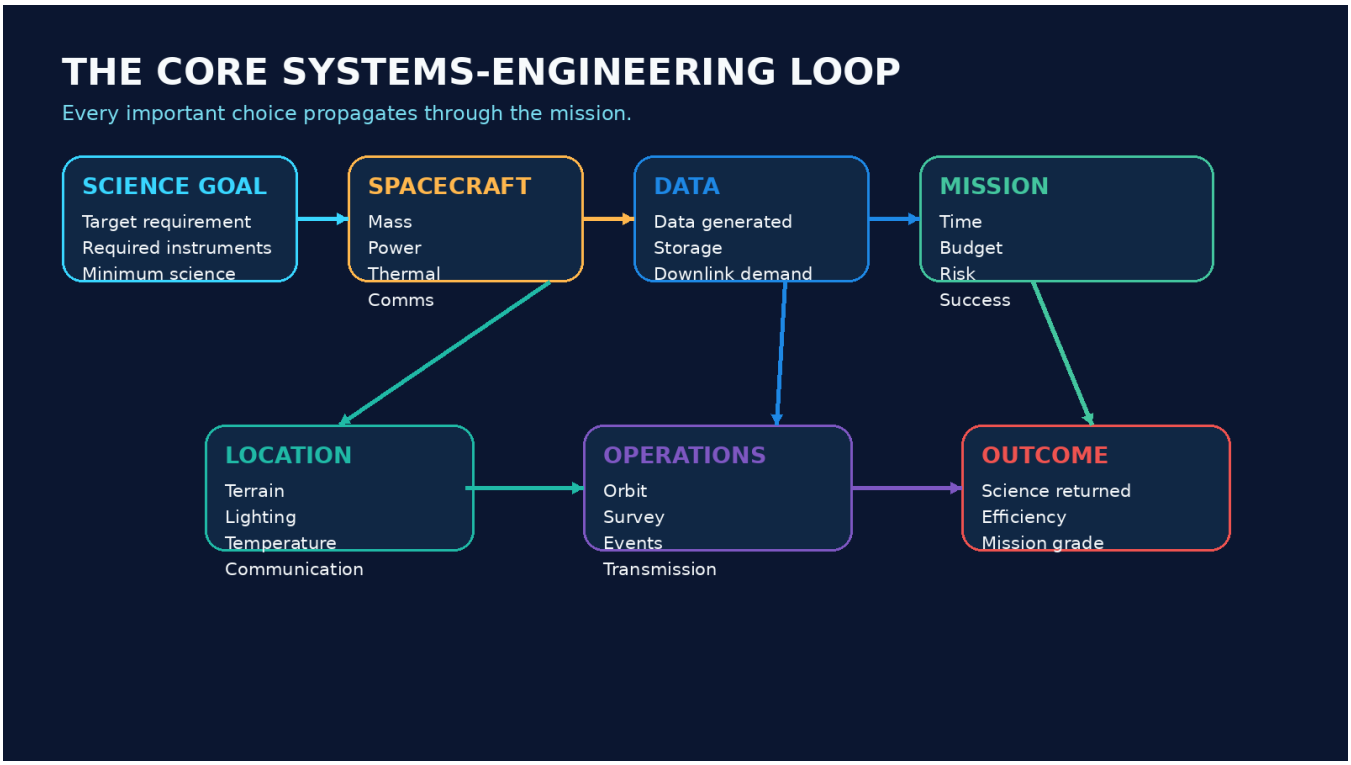
| Interaction | Meaning | Example consequence |
|---|---|---|
| Deploy solar array | Choose a larger or smaller power architecture. | Power +, mass +, cost + |
| Deploy antenna | Increase communication capability. | Bandwidth/data return +, mass/power + |
| Add thermal structure | Increase environmental tolerance. | Risk -, mass/cost + |
| Attach instrument | Increase scientific capability. | Science/data +, power/mass/cost - |
| Modify propulsion module | Increase transfer/descent margin. | Fuel margin +, mass/cost + |

**VISUAL GOAL**

The spacecraft should visibly transform as it is engineered, so the player can understand their architecture without reading a technical diagram.

# Mission Resource & Trade-off Model

The main mechanic is propagation: one decision changes several downstream variables.




| Primary constraint | Game meaning |
|---|---|
| Budget | Development + operational spending limit. |
| Mass | Launch and spacecraft capacity constraint. |
| Power | Available energy for systems and instruments. |
| Data | Scientific information generated and stored. |
| Communications | Ability to return data during available windows. |
| Time | Mission lifetime and operational window. |
| Science | Useful scientific return; mission objective score. |
| Risk | Derived condition affected by aggressive or incompatible choices. |

# Mission Simulation

Once the design is locked, the player experiences the mission as a sequence of operational decisions.

| Stage | Player sees | Player does |
|---|---|---|
| Design review | Spacecraft + resource telemetry | Fix incompatible design. |
| Launch | Animated launch sequence + vehicle status | Observe / proceed. |
| Transfer | Earth-Moon trajectory, time and fuel telemetry | Choose transfer/orbit strategy when appropriate. |
| Arrival | Chosen lunar site and environment | Confirm site conditions. |
| Science operations | Instrument activity, environment, power, data growth | Choose observation intensity and respond to events. |
| Downlink | Data volume + communication window | Prioritize, compress or transmit. |
| Result | Mission score + visual explanation | Understand cause and effect. |

**SIMULATION PRINCIPLE**

The simulation should be physically inspired but gameplay-sized. Use real science where it improves understanding; use transparent simplified models where full fidelity would add complexity without educational value.

# NASA Data & Scientific Layer

## The supplied lunar CSV is a first-class project input, not decoration.

The existing NASA CSV contains approximately 21 lunar/environmental values with units, usage notes and NASA source links. The game should preserve the source information and clearly distinguish measured values, estimates/uncertainties and gameplay approximations.

| NASA data family | Example use in game |
|---|---|
| Gravity | Simplified propulsion/orbit modifier and educational comparison. |
| Escape velocity | Mission difficulty / propulsion context. |
| Temperature | Thermal-system requirements and environment visuals. |
| Extreme cold | Environmental event / thermal risk. |
| Atmosphere / oxygen | Contextual science explanation; not a player resource. |
| Day / night duration | Power availability and operational planning. |
| Water / ice | Science-target value and observation motivation. |
| Radiation | Environmental risk and science context. |

**SCIENTIFIC HONESTY**

Never present a game-derived value as if it were directly measured by NASA. Label estimates and simplified game models clearly.

# Visual Mission Debrief

The final report is a major part of the product: it converts technical state into an understandable visual story.

| Report layer | Visual treatment |
| --- | --- |
| Mission route | Animated Earth -> Moon -> selected site path. |
| Actual spacecraft | Reconstruct the exact configuration the player engineered. |
| Environment | Visual temperature, illumination, terrain and communication indicators. |
| Science | Maps, diagrams, scan areas, charts and observation results. |
| Data | Generated vs returned volume, storage and downlink visualization. |
| Decision impact | Before/after bars, cause -> effect chains and short explanations. |
| Outcome | Success / partial / failure with a clear reason and next-step insight. |

The report must be understandable to a non-technical judge. The player should be able to point to a visual and say: "This choice caused this constraint, which changed this outcome."

# Visual Direction & Procedural Rendering

The target is high-fidelity 2D scientific visualization, not retro arcade styling.

**VISUAL TARGET**

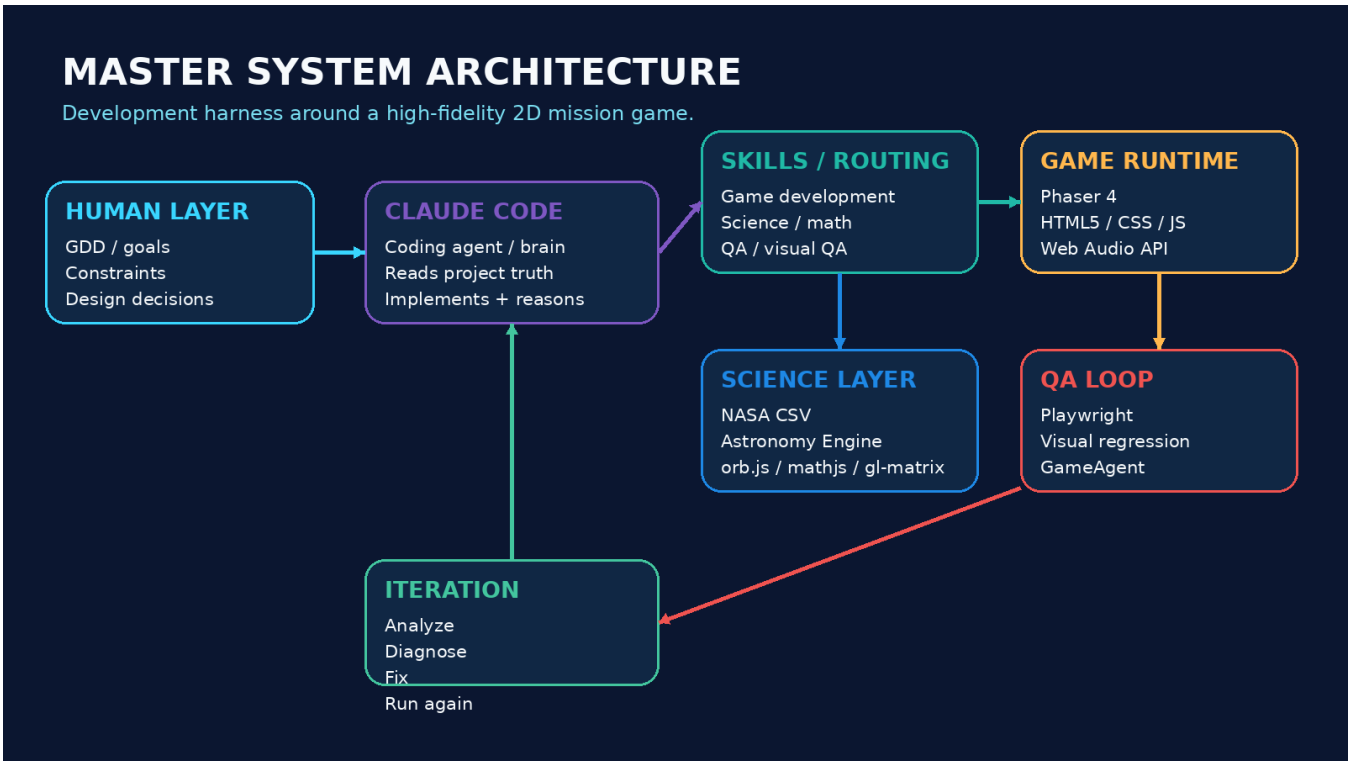
Cinematic, high-fidelity 2D space visualization: realistic-feeling celestial bodies, physically inspired lighting, detailed procedural spacecraft, smooth camera motion, restrained mission-control UI and sophisticated scientific graphics.

- No pixel art, 8-bit, CRT or deliberate arcade-retro treatment.
- Use Phaser 4 with a modern GPU rendering path for smooth 2D presentation.
- Generate Earth, Moon, planets, spacecraft, Origami geometry, stars, craters, effects and mission graphics procedurally.
- Use lighting, gradients, particles, masks, glows and camera movement to create depth.
- Preserve the mission-control HUD language from the previous prototype while substantially increasing world, spacecraft and scientific-report fidelity.

| Layer | Visual responsibility |
|---|---|
| World | Solar system, celestial bodies, lunar sites, environmental effects. |
| Craft | Procedural spacecraft geometry + animated deployment. |
| Motion | Camera travel, zoom, orbit, launch, scanning, downlink. |
| UI | Mission HUD, engineering panels, selection states, telemetry. |
| Science | Maps, charts, thermal/radiation/illumination visualizations. |
| Report | Visual mission reconstruction and decision analysis. |

# Runtime & Science Architecture

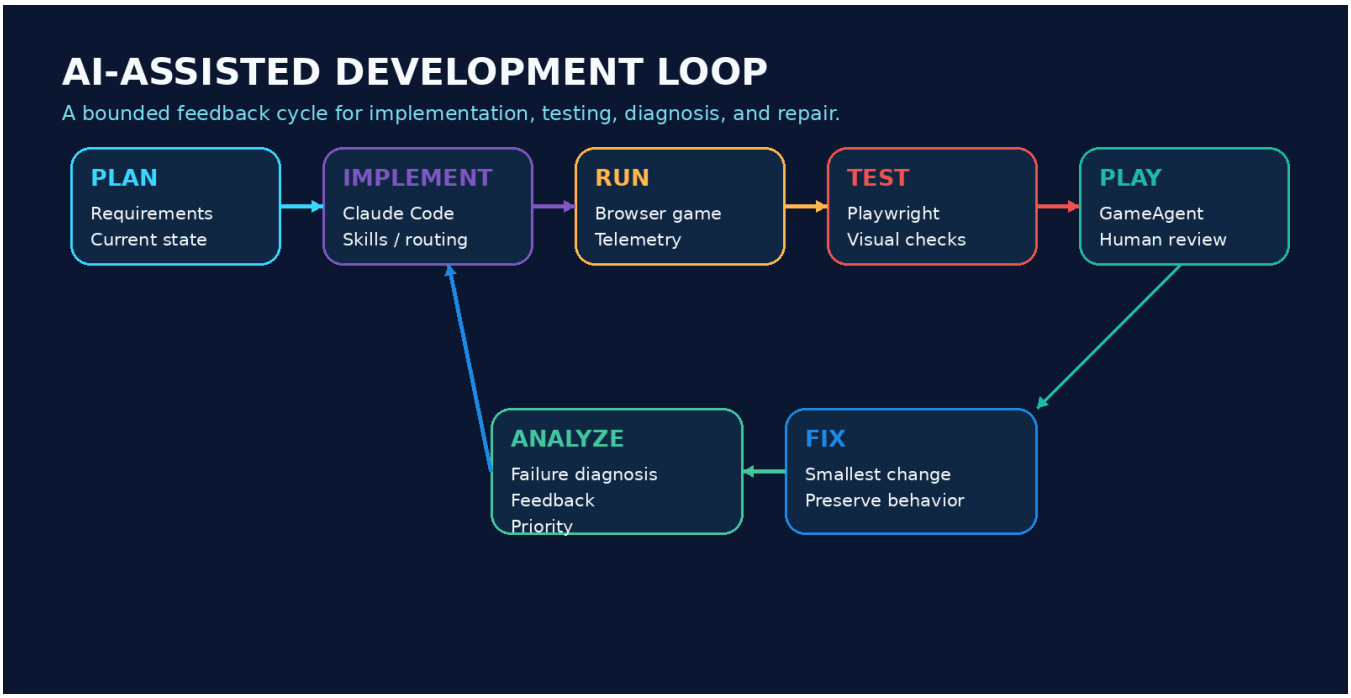
A focused stack keeps the game powerful without creating unnecessary infrastructure.




| Layer | Technology | Role |
|---|---|---|
| Runtime | HTML5 + CSS3 + Vanilla JS | Browser shell, styling, application logic. |
| 2D engine | Phaser 4 | Scenes, cameras, tweens, input, rendering, game runtime. |
| Audio | Web Audio API | Procedural BGM and sound effects. |
| Astronomy | Astronomy Engine | Sun/Moon/planet positions, timing, celestial calculations. |
| Orbital math | orb.js | Specialized orbital propagation when the mission model requires it. |
| General math | gl-matrix + mathjs | Vectors, matrices, transforms, equations, units and general engineering math. |
| Data | NASA CSV + structured local data | Scientific/environmental inputs. |

# AI-Assisted Development Harness

Claude Code is the central coding brain; skills and QA systems surround it.




The specialized skill files already created by the team remain the knowledge layer for the workflow. Tool routing decides when to use game-development knowledge, astronomy/orbital computation, visual systems, automated browser testing, visual regression or autonomous gameplay.

# Execution Roadmap

Build the smallest complete vertical slice first, then increase fidelity.

| Phase | Deliverable | Exit condition |
|---|---|---|
| 1. Foundation | Phaser 4 project, game loop, scene/state system, procedural render base. | Game boots and renders at target resolution. |
| 2. Exploration | Solar-system view, planet/moon interaction, smooth camera travel. | Player can explore and reach Moon. |
| 3. Mission setup | Objective selection, lunar site profiles, mission requirements. | A site creates visible requirements. |
| 4. Engineering | Origami spacecraft, subsystem choices, resource propagation. | A design can pass/fail validation. |
| 5. Simulation | Launch, transfer, orbit, survey, event, data generation. | Mission can be completed end-to-end. |
| 6. Downlink | Communication/data system + transmission result. | Science can be successfully or partially returned. |
| 7. Debrief | Visual report with NASA data and decision consequences. | Player can understand why result occurred. |
| 8. QA | Playwright + visual regression + GameAgent sessions. | Critical flows pass consistently. |
| 9. Polish | Lighting, motion, particles, typography, audio, transitions. | Experience looks cohesive and presentation-ready. |

# Quality Strategy & Definition of Done

Quality is part of the architecture, not a final-day activity.

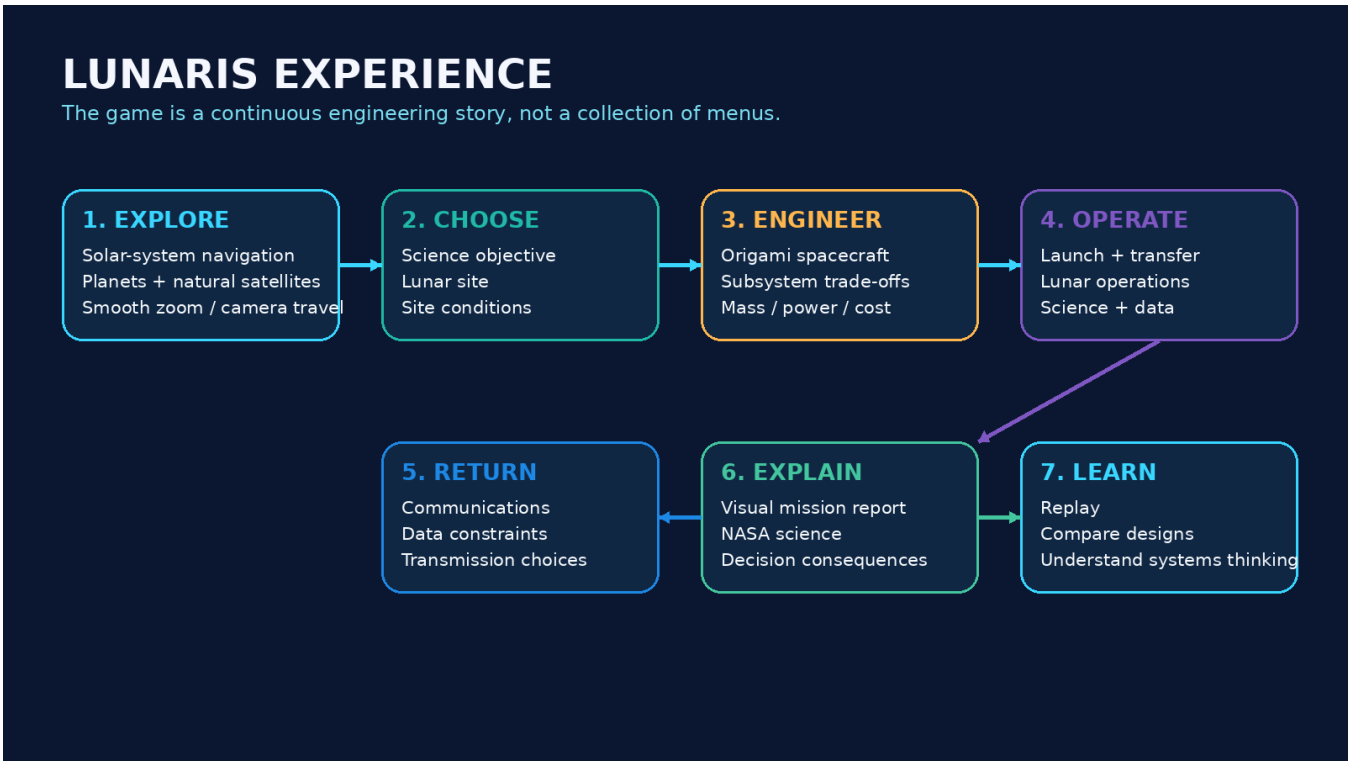
- Functional tests cover mission start, site selection, spacecraft configuration, validation, simulation, data transmission and report generation.
- Visual regression checks critical scenes: solar-system view, Moon zoom, spacecraft engineering, mission simulation, science survey and final report.
- GameAgent playthroughs search for dead ends, broken state transitions, impossible missions and confusing feedback.
- Scientific checks verify that displayed NASA values match the supplied dataset and that estimates are labelled.
- Performance checks protect smooth camera movement, animations and procedural rendering.

**DEFINITION OF DONE**

A player who knows nothing about mission design can complete a mission, understand the major decisions, see genuine NASA science represented visually, and explain why the final mission result happened.

# One-Page Blueprint

The whole project in one view.




SYSTEM MESSAGE

## CHOOSE THE MOON. ENGINEER THE SPACECRAFT. DISCOVER THE CONSEQUENCES.

| Product identity | Decision |
|---|---|
| Experience | High-fidelity 2D scientific visualization that happens to be a game. |
| Scope | One integrated lunar mission with deep systems interaction, not many shallow missions. |
| Signature mechanics | Solar-system exploration + lunar site selection + Origami spacecraft engineering + visual mission debrief. |
| Science | NASA lunar dataset is integrated into environment, requirements and debrief. |
| Technology | Phaser 4 + JS/HTML/CSS + Web Audio + astronomy/orbital/math skills. |
| AI workflow | Claude Code + specialized skills + orchestration + Playwright + Visual QA + GameAgent. |